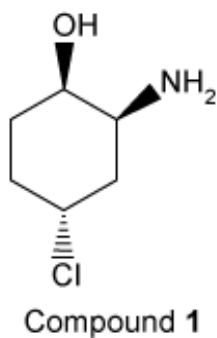
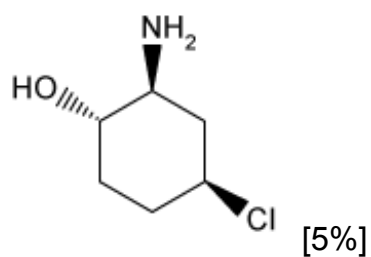


Compound 1 is shown below.

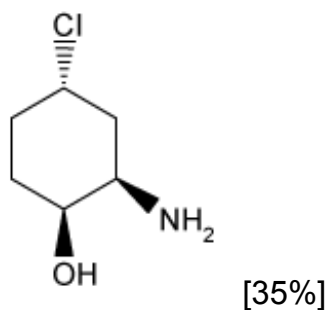


Which of the following structures is the enantiomer of Compound 1?

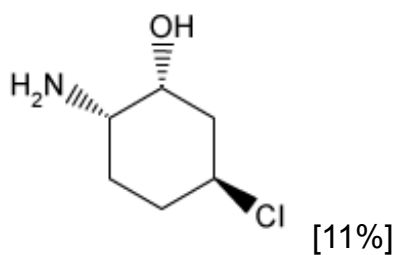
☐ A.



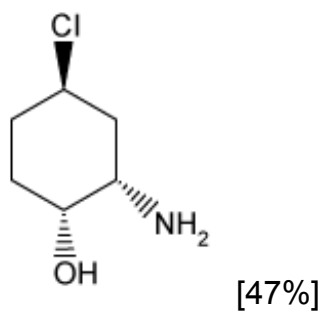
✓ ☒ B.



☐ C.



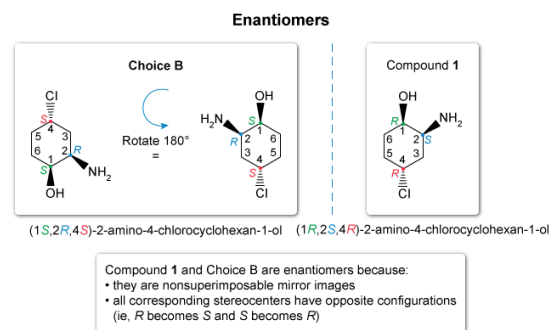
☐ D.



Correct

34% answered correctly

Explanation:



Stereoisomers are molecules with the same **molecular formula** and atom connectivity but different spatial arrangements. Stereoisomers contain one or more **stereocenters**, also known as **chiral centers**. The **priority** and circular arrangement of the substituents around the chiral center determines the ***R* or *S* configuration** of the stereocenter. **Enantiomers** are nonsuperimposable mirror images containing one or more stereocenters in which **all corresponding stereocenters are opposite in configuration**.

Compound **1** has chiral centers at carbons 1, 2, and 4. Based on the substituent priority, these centers have *R*, *S*, and *R* configurations, respectively. Rotating the structure shown in Choice B by 180° shows that it is a nonsuperimposable mirror image of Compound **1**. The configurations of the chiral centers at carbons 1, 2, and 4 in Choice B are opposite those in Compound **1** (ie, *S*, *R*, and *S*, respectively). Therefore, the structure in Choice B is the enantiomer of Compound **1**.

(Choice A) In this structure, all the stereocenters have a configuration of *S*. Because the configuration is inverted at only 2 of the 3 stereocenters, this structure is a **diastereomer** of Compound **1**.

(Choice C) This structure contains the same **molecular formula** as Compound **1** but has **different connectivity** (ie, Cl is bonded to carbon 5 instead of carbon 4). Therefore, this structure is a **constitutional isomer** (structural isomer) of Compound **1** rather than an enantiomer.

(Choice D) This structure has the **same connectivity and stereocenter configuration** as Compound **1**. Therefore, this structure is the **same** molecule as Compound **1** but shown in a different orientation, and cannot be an enantiomer.

Educational objective:

Stereoisomers are molecules with the same molecular formula and atom connectivity but different spatial arrangement. Enantiomers are stereoisomers that are nonsuperimposable mirror images containing one or more stereocenters, and with all the corresponding stereocenters opposite in configuration.

Time spent: 0 sec

Subject:
Organic Chemistry

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System:
Introduction to Organic Chemistry